CURRICULUM VITAE

Sishi Li (Dr)

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CURRENT POSITION

10/2025 – Present Research Fellow, Department of The Built Environment, National University of Singapore, Singapore

EDUCATION

08/2020 – 06/2025 **Ph.D.**
Tsinghua University, Beijing, 100084, China

- Civil Engineering, Department of Building Science, School of Architecture
- Supervisor: Prof. Bin Cao
- Dissertation: Design and evaluation of a dynamic non-uniform thermal environment
- GPA: 3.87/4.00
- Main courses: Advanced Heat & Mass Transfer; Thermal Principle of Indoor Built Environment; Building Indoor Thermal Environment Simulation & Application; Methods of Optimization, etc.

09/2016 – 06/2020 **B.Eng.**
Chongqing University, Chongqing, 400044, China

- Building Environment and Energy Engineering, School of Civil Engineering
- GPA: 3.84/4.00 (rank: 1/114)
- Main courses: Engineering Thermodynamics; Heat transfer; Fluid Mechanics; Principles and Equipment of Heat & Mass Transfer; Built Environment; Cooling & Heating Sources for Buildings; Heating, Ventilation & Air Conditioning, etc.

RESEARCH INTERESTS

- General: Built Environment, Thermal comfort, Thermal protection
- Specific:
  - Dynamic thermal environment design (e.g., indoor dynamic temperature control)
  - Personal thermal comfort (e.g., operation strategies and effectiveness evaluation of personal environment comfort system)
  - Thermal protection in extremely cold environments (e.g., thermal protection against extreme cold exposure and large temperature drops)

SKILLS

Software
- Data analysis: R, MATLAB, Python, SPSS
- Drawing/Modeling: AutoCAD
- Others: Microsoft Office, DeST

Experiments
- Field tests: Field environment measurement, HVAC system performance tests, Filed survey design
- Climate chamber experiments: Subjective experiments & Thermal manikin tests

HONORS & AWARDS

2024 Second prize of China Thermal Comfort Conference for Student Frum

2023	First-class Scholarship of Tsinghua University
2021 & 2024	Second-class Scholarship of Tsinghua University
2020	Excellent Graduate of Chongqing University
2019	National Encouragement Scholarship of China for Undergraduate Students
2017	National Scholarship of China for Undergraduate Students
2017 – 2019	First-class Excellent Student Scholarship of Chongqing University

RESEARCH PROJECT EXPERIENCES

Green Buildings Innovation Cluster (GBIC) R&D (PI: Ali Ghahramani) 9/2025 – Now

- Research: Digital twin-based smart fan systems for enhanced energy efficiency and occupant comfort
- Main work: Design the control logic for the smart fan system; simulation validation of the fan system's performance; Project management works.
- Role: Leading Researcher

National Key Research and Development Program of China (PI: Bin Cao) 11/2022 – 11/2025

- Research: Personal environmental comfort demand analysis and system design in non-uniform space-time environments
- Main work: Analyze the spatial and temporal characteristics of environmental parameters and human needs; quantify the effects of typical personal environmental comfort systems (PECS) on human thermal comfort and building operational energy consumption; and develop a support platform to guide the design and operation of PECS.
- Role: Leading Researcher and Project Management of Sub-project

Tsinghua University-Toyota Joint Research Fund (PI: Bin Cao) 11/2022 – 11/2024

- Research: Humanized design and products for the user interface in the built environment
- Main work: Analyze the effects of multiple non-uniform heat and cold exposures on human physiological responses and subjective feedback; propose a method for evaluating human thermal comfort in dynamic and non-uniform thermal environments.
- Role: Leading Researcher and Project Management of Sub-project

National Key Research and Development Program of China (PI: Bin Cao) 08/2018 – 12/2021

- Research: Therma protection technology for winter outdoor spectating environment
- Main work: Analyze the heating demand of the human body in extremely cold environments under different activity levels; develop a series of portable and wearable body heating devices applied to provide heat protection for staff and spectators at winter sports events.
- Role: Primary Researcher

College Student Research Training Program of Chongqing 06/2018 – 06/2019

- Research: Optimization design and indoor comfort analysis of deflecting ventilation
- Main work: Design and parameter optimization of deflector plate using numerical simulation; practical creation and experimental validation of deflection ventilation.
- Role: PI

PUBLICATIONS

[1] **S. Li**, X. Jia, B. Cao, et al. Thermal comfort and physiological responses under winter temperature cycles: Implications for demand response control in buildings, 2026 (Under Review of *Building and Environment*)

[2] **S. Li**, X. Jia, C. Peng, et al. Thermal comfort benefits, energy efficiency, and occupant regulation behaviour in four types of personal heating within the built environment, *Building and Environment* 266 (2024) 112075.

[3] **S. Li**, X. Jia, C. Peng, et al. Effects of temperature cycles on human thermal comfort in built environment under summer conditions, *Science of The Total Environment* 912 (2024) 168756.

[4] **S. Li**, X. Jia, B. Cao, et al. Improvement of thermal perceptions and physiological responses using torso heating under short-term cold exposure, *Science of The Total Environment* 892 (2023) 164491.

[5] **S. Li,** X. Jia, B. Cao, et al. Thermal comfort characteristics and heating demand of people with different activity status during extremely cold exposure, *Building and Environment* 228 (2023) 109798.

[6] **S. Li,** Y. Deng, B. Cao, Study on the Performance of Personal Heating in Extremely Cold Environments Using a Thermal Manikin, *Buildings* 13 (2023) 362.

[7] X. Jia, **S. Li**, Y. Zhu, et al. Development of the Universal Standard Effective Temperature for evaluating thermal comfort across different metabolic rates, *Building and Environment* 250 (2024) 111149.

[8] X. Jia, **S. Li**, J. Wang, et al. Measurement of steady state and dynamic metabolic rates under typical daily activities, *Building and Environment* 254 (2024) 111344.

[9] X. Jia, **S. Li**, Y. Zhu, et al. Transient thermal comfort and physiological responses following a step change in activity status under summer indoor environments, *Energy and Buildings* 285 (2023) 112918.

[10] J. Zhao, R. Wang, Y. Wu, ……, **S. Li**, H. Jiang, B. Cao, From characteristics to practical applications of skin temperature in thermal comfort research – A comprehensive review, *Building and Environment* 262 (2024) 111820.

[11] **S. Li**, B. Cao, Study on Wearable Personalized Thermal Comfort System for Cold Environments, *World Architecture* 6 (2022) 61-65. (in Chinese)

[12] **S. Li**, M. Zhao, Y. Zhai, et al. Optimization Study of Deflecting Ventilation Based on CFD and Orthogonal Tests, *Building Energy & Environment* 8 (2020) 25-30. (in Chinese)

[13] B. Cao, Y. Zhu, Y. Hou, ……, **S. Li**, S. Liu, Ergonomics in built environments: Prospects of human thermal comfort research, *Chinese Science Bulletin* 67 (2022) 1757-1770. (in Chinese)

CONFERENCES

[1] **S. Li**, B. Cao, Y. Zhu. The effects of temperature cycles on thermal comfort in built environment under summer conditions. *Thermal Comfort Conference and the 5th Academic Forum of the Thermal Comfort Group and the Sleep Environment and Health Group in China*, November 15-17, 2024. (oral)

[2] **S. Li**, B. Cao, Y. Zhu. Thermal performance evaluation of four personal heating devices and occupant usage behaviour under three cold conditions. *Indoor Air 2024*, July 7-11, 2024, Honolulu, USA. (oral)

[3] **S. Li**, B. Cao, Y. Zhu. Human responses to temperature cycles under summer conditions. *The 13th International Symposium on Heating, Ventilation, and Air Conditioning (ISHVAC 2023)*, November 14-16, 2023, Beijing, China. (oral)

[4] **S. Li**, B. Cao. Thermal perception and physiological response of people with different heating modes during short-term cold exposure. *Healthy Buildings 2023 Asia and Pacific Rim*, July 17-19, 2023, Tianjin, China. (oral)

[5] **S. Li**, B. Cao, Y. Zhu. Effect of activity status on thermal comfort during extremely cold exposure. *Indoor Air 2022*, June 12-16, 2022, Kuopio, Finland. (online; poster)

TEACHING & MENTORING EXPERIENCES

2024	Peidong Lin, Sichen Zhen, undergraduate research, the thermal perception and physiological responses of people under dynamic temperature changes.
2022 – 2023	Huanyu Yue, Undergraduate Thesis, Local distribution of human thermal sensitivity and thermal comfort temperature threshold.

2022 Teaching Assistant for the undergraduate course "Built Environment", three credits, 25 undergraduate students, Tsinghua University. Responsibilities include:

- Preparing new course materials, managing the online course, and answering students' questions before the class
- Conducting pre-course tests and teaching them after tests; facilitating the course discussion sessions; designing the teaching experiments
- Grading assignments, final exams, course research training programs, experimental reports, and group presentations

2021 Teaching Assistant for the undergraduate course "Indoor Environment & Health", one credit, 36 undergraduate students, Tsinghua University. Responsibilities include:

- Preparing new course materials and organizing off-campus study tours
- Arranging mid-term and final tasks and group presentations

PROFESSIONAL SERVICE

- **Journal peer reviewer**: Building and Environment, Energy and Buildings, Building Simulation

LIST OF REFERENCES

Bin Cao, Ph.D.
Associate Professor, Department of Building Science, School of Architecture, Tsinghua University, Beijing, China
Vice head of the Department of Building Science, Tsinghua University
Chair of STC 33 (Thermal Comfort) of the International Society of Indoor Air Quality
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